

Module 2: Satellite Link Budget Analysis and Atmospheric Loss

The satellite communication link budget calculates received carrier power using Friis transmission equation. Equivalent Isotropically Radiated Power EIRP represents the product of transmit power and antenna gain. Path loss and rain attenuation degrade downlink signal strength at higher microwave frequencies. Carrier to noise ratio CNR and system noise temperature determine receiver sensitivity and bit error rate. Link margin calculations account for atmospheric absorption, antenna pointing errors, and ionospheric scintillation.